

API Research Report

Miscellaneous / Utility APIs — Intern 8

Evaluated APIs: OpenAlex | EventRegistry | RapidAPI Real-Time News Data

Prepared for Internship Presentation

Date: July 24, 2026

This report summarizes hands-on research and testing of three APIs, covering authentication, data formats, free-tier limitations, integration effort, and recommended use cases — including current-as-of-2026 verification of key details.

Objective

The objective of this task was to research, test, and evaluate three APIs — **OpenAlex**, **EventRegistry**, and a **RapidAPI** service (Real-Time News Data). The evaluation focused on the data each API provides, its authentication method, free-tier limitations, ease of integration, and overall suitability for project use.

1. OpenAlex API

API Link: <https://api.openalex.org/>

Purpose: OpenAlex is a free, open scholarly database that indexes hundreds of millions of research papers, authors, institutions, journals, and citation relationships.

Data Provided

- Research paper titles, abstracts, and full metadata
- Authors and their institutional affiliations
- Publication year, DOI, and citation counts
- Research topics, concepts, and keywords
- Referenced works and related works

Authentication & Access — Important 2026 Update

During earlier testing, OpenAlex required no API key and was fully open to anonymous requests (the "polite pool" email parameter simply improved response speed). **This has changed.** Starting February 2026, OpenAlex introduced mandatory API keys and a usage-based pricing model; the polite-pool email parameter was retired. Anonymous requests without a key are now limited to a small free credit allowance before returning 409 errors. It remains free for most typical academic use, but teams should register for a key rather than relying on the older no-auth behavior.

Prior/base rate limits (still relevant as a ceiling under the free allowance): up to 100,000 calls per day and 10 requests per second.

Data Format

JSON

Testing Summary

- A Python script using the **requests** library queried papers related to Artificial Intelligence
- API call successful — Status Code: **200 OK**
- Returned detailed JSON metadata for matching research papers
- Output saved as **openalex_output.json**

Challenges

- Very detailed JSON responses that require filtering to extract useful fields
- Large responses can need extra processing/pagination for bigger queries
- New key requirement (2026) means existing no-auth scripts must be updated

Advantages

- Free for most standard use, even after the 2026 policy change
- Reliable, well-documented, and actively maintained
- Very rich, structured academic metadata

Verdict: Highly Suitable

Best choice for academic research, citation analysis, and scientific-publication use cases — provided a registered API key is used going forward.

2. EventRegistry API

API Link: <https://eventregistry.org/>

Purpose: EventRegistry is a news-intelligence platform that aggregates and analyzes news articles from thousands of global sources, clustering related coverage into detected "events."

Data Provided

- News articles and breaking news
- Detected global events and event clusters
- Topics, categories, and locations
- Organizations mentioned in coverage
- Event-level analytics and sentiment

Authentication

Requires an API key, obtained after creating a free account.

Data Format

JSON

Free Tier

A free account is available with a limited monthly token/request allowance. Advanced features such as deeper historical search, sentiment analysis, and higher volume access sit behind paid subscription tiers.

Testing Summary

- Free account created and verified; API key obtained
- Python script successfully retrieved news articles in JSON format
- Output saved successfully

Challenges

- The API key/account setup flow is not immediately obvious for new users
- Several advanced endpoints are gated behind paid plans
- Documentation is more complex to navigate than OpenAlex's

Advantages

- High-quality, structured news data with event clustering
- Useful filtering by topic, location, and organization
- Good fit for monitoring and trend-detection use cases

Verdict: Partially Suitable

Useful for news-monitoring and event-tracking projects, but free-tier request/token limits restrict access to some advanced features and should be budgeted for in advance.

3. RapidAPI — Real-Time News Data API

Marketplace Link: <https://rapidapi.com/>

Purpose: RapidAPI is an API marketplace hosting thousands of third-party APIs under one platform and billing system. For this task, the "Real-Time News Data" API (a third-party listing on the marketplace) was selected and tested.

Data Provided

- Latest news headlines and full story coverage
- News categories and publisher information
- Article links and publication timestamps

Authentication

A RapidAPI account and a RapidAPI key are required; the key is passed in request headers and is shared across all APIs subscribed to on the platform.

Data Format

JSON

Free Tier

A free subscription tier is available for most listings, but monthly request limits vary by the individual API provider — not by RapidAPI itself. Each API on the marketplace sets its own pricing and quota structure.

Testing Summary

- Tested using the Full Story Coverage endpoint
- RapidAPI's auto-generated request code was executed using Python
- API call successful — Status Code: **200 OK**
- Output saved as **rapidapi_output.json**

Challenges

- A subscription step is required before any endpoint can be called
- Request limits differ across individual APIs on the marketplace
- The interface can feel unfamiliar to first-time users

Advantages

- Fast integration via auto-generated code snippets in multiple languages
- One platform, one key, access to a huge range of third-party APIs
- Convenient for prototyping when comparing multiple data providers

Verdict: Suitable

A good option for live news data, especially when quick integration matters. Free-tier request limits should be monitored per underlying provider.

Comparison Table

Feature	OpenAlex	EventRegistry	RapidAPI (Real-Time News Data)
Data Type	Academic research	News & events	Live news
Authentication	API key (required since Feb 2026)	API Key	RapidAPI Key
Data Format	JSON	JSON	JSON
Free Tier	Yes (usage-based after Feb 2026)	Limited	Limited (varies by provider)
Python Support	Yes	Yes	Yes
Ease of Use	Very Easy	Moderate	Easy
Test Status Code	200	200	200
Tested Successfully	Yes	Yes	Yes

Note: The "Authentication" and "Free Tier" rows for OpenAlex reflect a platform policy change introduced in February 2026 (mandatory API keys, usage-based pricing), which supersedes the fully open, no-key access available at the time of the original testing.

Challenges Faced

OpenAlex

- Large and complex JSON responses that need filtering
- Existing no-auth scripts require updating for the new key requirement

EventRegistry

- API key generation/setup was not immediately obvious
- Free account has restricted access to advanced features

RapidAPI

- Understanding the marketplace interface took extra time
- A subscription step is required before accessing any endpoint
- Free-tier limits vary per underlying API, not per RapidAPI itself

Recommendation

OpenAlex — Best choice for academic research, scientific publications, and citation analysis; register for an API key to stay within the current policy.

EventRegistry — Recommended for news intelligence and event tracking; plan around free-tier request limits.

RapidAPI — Recommended when multiple different APIs are needed from a single platform, thanks to easy integration and multi-language code generation.

Conclusion

All three APIs were successfully researched and tested using Python. OpenAlex proved the easiest to use thanks to its open academic data, though it now requires a free API key following its February 2026 policy update. EventRegistry provided rich news intelligence but requires authentication and carries free-tier limitations. RapidAPI served as a convenient marketplace for testing third-party APIs, offering fast integration through auto-generated code snippets.

Overall, each API suits a different use case, and the right choice depends on the specific requirements of the project — academic data, news intelligence, or marketplace flexibility.

Short Presentation Script (2–3 Minutes)

For this task, I researched and tested three APIs: OpenAlex, EventRegistry, and RapidAPI's Real-Time News Data API. I created accounts where necessary, obtained API keys, and used Python with the requests library to make real API calls.

OpenAlex provides academic research data. At the time of my testing it needed no authentication, though I should note that OpenAlex introduced a mandatory API-key and usage-based pricing policy in February 2026, so any production use today should register for a key. It returned detailed metadata about research papers in JSON format.

EventRegistry provides global news and event intelligence. It requires an API key and offers structured news data, although some advanced features are limited on the free plan.

For RapidAPI, I tested the Real-Time News Data API. After subscribing to the free plan, I successfully retrieved live news data using Python. RapidAPI also provides auto-generated code snippets, making integration faster.

After comparing all three, I found OpenAlex best for academic research, EventRegistry suitable for news analysis, and RapidAPI a flexible platform for accessing many different APIs. All three successfully returned JSON data and can be integrated into future applications depending on project requirements.